

Introductory Econometrics with Recitation — Quiz 5

May 19, 2026

Answer Key

1. (17 points) The following table reports two panel regressions for 50 cities from 2017 to 2023. Each observation is a city-year, and all variables in this table are measured at the city-year level. The variables are defined as follows:

Variable Name	Description
<i>InjuryRate</i>	Number of e-scooter injuries per 100,000 residents in a city-year.
<i>HelmetFine</i>	Fine amount, in dollars, for riding an e-scooter without a helmet in that city-year.
<i>Age18</i>	= 1 if the city's minimum riding age is 18 in that year, and 0 otherwise.

Table 1: Impact of Helmet Fines on E-Scooter Injuries

Regressor	Dependent variable: <i>InjuryRate</i>	
	(1)	(2)
$\ln(\text{HelmetFine})$	-1.80 (0.55)	-1.60 (0.50)
<i>Age18</i>	-0.52 (0.22)	-0.46 (0.21)
City Fixed Effects	Yes	Yes
Year Fixed Effects	No	Yes
Clustered Standard Errors	Yes	Yes
Observations	350	350
R^2	0.68	0.72

Notes: Standard errors are reported in parentheses.

- (a) (8 points) Using Model (1), test whether the coefficient on $\ln(\text{HelmetFine})$ is statistically significant at the 5% level. Interpret the coefficient in words.
- (b) (3 points) Why might the regression include city fixed effects? Give one example of an omitted variable that can be captured by city fixed effects.
- (c) (3 points) Why might Model (2) include year fixed effects? Give one example of an omitted variable that can be captured by year fixed effects.
- (d) (3 points) How should standard errors be clustered in this regression? Explain briefly.

Suggested Answers for Problem 1:

- (a) Let β_1 be the coefficient on $\ln(\text{HelmetFine})$ in Model (1). The hypotheses are

$$H_0 : \beta_1 = 0, \quad H_A : \beta_1 \neq 0.$$

The test statistic is

$$t = \frac{-1.80}{0.55} \approx -3.27.$$

Since $|t| = 3.27 > 1.96$, we reject H_0 at the 5% level. The coefficient is statistically significant and negative.

Because the dependent variable is measured as injuries per 100,000 residents and the regressor is $\ln(\text{HelmetFine})$, a 1% increase in the helmet fine is associated with about

$$\frac{-1.80}{100} = -0.018$$

fewer e-scooter injuries per 100,000 residents, holding *Age18* and city fixed effects constant.

Grading Criterion

- **(1 pts)**: Correctly states the null and alternative hypotheses.
- **(3 pts)**: Correctly computes the test statistic.
- **(1 pts)**: Correct conclusion at the 5% level.
- **(3 pts)**: Correct interpretation of the log coefficient.

- (b) City fixed effects control for time-invariant differences across cities. These are omitted variables that differ across cities but do not change much over time. Examples include road design, long-run traffic culture, baseline scooter usage, or persistent differences in enforcement intensity.

Grading Criterion

- **(2 pts)**: Correctly explains that city fixed effects control for time-invariant city-level omitted variables.
- **(1 pt)**: Gives a reasonable city-level example.
- **(0 pts)**: Incorrect answer.

- (c) Year fixed effects control for shocks or omitted variables that affect all cities in the same year. Examples include national safety campaigns, national changes in e-scooter technology, national economic conditions, or national changes in traffic behavior.

Grading Criterion

- **(2 pts)**: Correctly explains that year fixed effects control for common year shocks.
- **(1 pt)**: Gives a reasonable year-level example.
- **(0 pts)**: Incorrect answer.

- (d) Standard errors should be clustered at the city level. The data are panel data, so observations from the same city over time may have correlated errors. Clustering by city allows arbitrary correlation within each city over time.

Grading Criterion

- **(3 pts)**: Correctly states city-level clustering and explains why.
- **(1–2 pts)**: Partially correct answers.
- **(0 pts)**: Incorrect answer.

2. **(13 points)** A researcher uses data on 700 private-sector employees to study whether employees are allowed to work remotely at least two days per week. The variables are defined as follows:

Variable Name	Description
$Remote_i$	= 1 if employee i is allowed to work remotely at least two days per week, and 0 otherwise.
$Experience_i$	Years of work experience for employee i .
$Female_i$	= 1 if employee i is female, and 0 otherwise.

Table 2: Regression Models for Remote Work Eligibility

Regressor	Dependent variable: $Remote$		
	Probit	Logit	LPM
Experience	0.120 (0.030)	0.200 (0.050)	0.025 (0.007)
Female	0.450 (0.280)	0.780 (0.490)	0.090 (0.060)
Constant	-1.250 (0.270)	-2.100 (0.470)	0.280 (0.060)
Observations	700	700	700

Notes: Standard errors are reported in parentheses.

- (a) **(3 points)** Using the probit model, compute the predicted probability of $Remote = 1$ for Liam, a man with 10 years of experience.
- (b) **(3 points)** Using the logit model, compute the predicted probability for Liam.
- (c) **(3 points)** Using the LPM, compute the predicted probability for Liam.
- (d) **(4 points)** Compare the linear probability model (LPM) with the probit model. Provide one advantage and one disadvantage of using the LPM.

Suggested Answers for Problem 2:

- (a) For Liam, $Female = 0$ and $Experience = 10$. The probit index is

$$-1.250 + 0.120(10) = -0.050.$$

Thus,

$$\hat{P}(Remote = 1) = \Phi(-0.050) \approx 0.480.$$

Grading Criterion

- **(3 pts)**: Correct predicted probability.
- **(1-2 pts)**: Partially correct answer.
- **(0 pts)**: Incorrect answer.

- (b) For Liam, $Female = 0$ and $Experience = 10$. The logit index is

$$-2.100 + 0.200(10) = -0.100.$$

Thus,

$$\hat{P}(Remote = 1) = \frac{1}{1 + e^{-(-0.100)}} = \frac{1}{1 + e^{0.100}} \approx 0.475.$$

Grading Criterion

- **(3 pts)**: Correct predicted probability.
- **(1-2 pts)**: Partially correct answer.
- **(0 pts)**: Incorrect answer.

- (c) For the LPM, the predicted probability for Liam is

$$0.280 + 0.025(10) = 0.530.$$

Thus, the predicted probability is 0.530, or 53.0%.

Grading Criterion

- **(3 pts)**: Correct predicted probability.
- **(1-2 pts)**: Partially correct answer.
- **(0 pts)**: Incorrect answer.

- (d) • **Advantage:** It is easy to interpret as the coefficients directly reflect changes in probability.

- **Disadvantage:** It can predict probabilities outside the $[0,1]$ range and assumes constant marginal effects, which may be unrealistic for binary outcomes.

Grading Criterion

- **(2 pts):** Provides one correct advantage of the LPM.
- **(2 pts):** Provides one correct disadvantage of the LPM.